

Paper BIO-01: Adaptive Flux Limitation in Living Systems: A CDFD Model for Biological Throughput

Steve Bico Mujjabi
Independent Researcher

May 22, 2026

Abstract

This paper presents **Adaptive Flux Limitation (AFL)** as the Part III biology-and-medicine model inside Constraint-Driven Flux Dynamics (CDFD). CDFD supplies the flow-constraint-memory grammar: physiological flow Φ , constraint C , surface responsiveness S , and structural memory M_s . The Runtime citation anchors the CDFL implementation, while clinical interpretation remains separate from software output and bedside validation.

Biological systems exchange energy, matter, and information under continuous throughput. Models that emphasize concentrations and molecular catalogs alone often under-represent saturation, bottlenecks, and adaptive regulation seen across physiology and disease.

We introduce **Adaptive Flux Limitation (AFL)**: living systems dynamically adjust constraints that bound throughput in order to remain in a regulable regime across an active, memory-bearing biological surface. We align AFL with the extended CDFD convention in which **flow intensity** is denoted Φ , **constraint** by C , **surface responsiveness** by S , and **surface memory** by M_s . The operating regime is summarized by the adaptive ratio $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. A minimal spatial toy model illustrates how biological surfaces dynamically integrate load, and how conditions such as disease emerge as surface dysregulation when memory (M_s) locks tissues into chronic constraint states.

Status: Inferred

Within the extended framework of this series, biological networks are modeled as *Adaptive Surfaces* that dynamically integrate physiological load and retain structural memory. The system's health is governed by the adaptive ratio $\Psi_s = (\Phi/C) \cdot S \cdot M_s$, where homeostasis ($\Psi_s \approx 1.0$) is maintained near the stable attractor ($S \cdot M_s \approx 1.0$), and disease emerges as surface dysregulation when maladaptive memory locks tissues into chronic constraint.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α , β , and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η , λ , μ , ν , ρ , σ , τ , and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Clinical Reader's Guide

This opening paper should be read as the dictionary for the medical series. The central move is simple: a living system is not only a collection of molecules, but a regulated surface that must move substrates, signals, heat, charge, cells, and information without losing structure. In that language, Φ is the demand placed on the system, C is the resistance or capacity burden that limits that demand, S is the ability of the tissue to respond, and M_s is the retained history of prior stress.

For a clinician, the useful question is not whether Ψ_s is a ready-made diagnostic number. It is not. The useful question is whether a bedside or laboratory pattern is behaving like a flux problem, a constraint problem, a responsiveness problem, or a memory problem. The same patient can contain several of these at once: anaemia can be limited by iron availability, inflammatory blockade, marrow capacity, renal signalling, or vascular tolerance. AFL is meant to keep those axes visible instead of collapsing them into one label.

The paper therefore distinguishes three layers. The first layer is notation and bookkeeping. The second is a toy dynamical model that shows how constraint can rise under sustained load. The third is the clinical hypothesis that similar geometry may appear in metabolic, renal, immune, oncological, developmental, and pharmacological systems. Only the first two are internal to the paper; the third must be tested with ordinary biomedical evidence.

1 Introduction

Adaptive Flux Limitation (AFL) is the named model used in this paper; CDFD is the parent framework that supplies the variables, closure logic, and claim discipline. The biological or medical question is posed first as AFL, then interpreted as a CDFD model of flow, constraint, surface responsiveness, and structural memory.

1.1 Formal Symbol Rigor: The Extended CDFD Paradigm

In strict alignment with the Fundamental Physics (Part I) and Origins of Life (Part II) archives, this biological translation formally preserves the exact Mujjabi transport tensor structure:

- Φ (**Flow Intensity**): The physiological or biochemical drive (e.g., nutrient flux, signaling rate).
- C (**Constraint / Pathway Resistance**): The biological resistance regulating the flow. It dynamically evolves via the standard closure: $dC/dt = \alpha\Phi - \beta C$.
- S (**Surface Responsiveness**): The active response capability of the biological medium.
- M_s (**Surface Memory**): Structural hysteresis or maladaptive drift (e.g., chronic scarring, epigenetic locking) with a finite recovery time τ_M .
- $\Psi_s = (\Phi/C) \cdot S \cdot M_s$ (**Operating State Ratio**): The dimensionless index of biological health. $\Psi_s \approx 1$ defines homeostasis ($\Psi_s \approx 1.0$); $\Psi_s \gg 1$ defines pathological overload.

Biological function is inherently dynamic: cells, tissues, and organisms transport substrates, signals, and heat. Across domains, similar qualitative patterns recur: response saturation at high input, bottleneck formation under stress, redistribution among parallel pathways, and stabilization near operating thresholds.

We propose that many such patterns are compatible with a single organizing picture: **throughput limited by constraints that themselves adapt to throughput**. This paper (AFL-1) states definitions, a compact dynamical model, and numerical verification for one illustrative implementation. It closes by mapping methodological alignment with the physics manuscript track without claiming new fundamental constants.

Open gap for AFL-2: whether the same mathematical skeleton appears across disparate biological domains with only reinterpretation of (Φ, C) .

Cross-link (multi-channel bioenergetics): When several transport modes co-limit throughput, the same public bookkeeping can be written as $J = \Phi \sigma_e \sigma_p / S$ with capacities $C_{\text{input}}, C_{\text{electron}}, C_{\text{proton}},$ and $C_{\text{stability}}$. In later papers this bridge is used only as a functional analogy: energy capture, electron transport, proton coupling, and stabilization may be carried by different biomaterials in different biological settings.

Status: Hypothesis

2 Model and Assumptions

2.1 Biological Translation of the Mujjabi Capacity Law [Mujjabi \[2026\]](#)

In Part I, the Mujjabi Capacity Law [Mujjabi \[2026\]](#) dictates that local transport becomes non-linear as the flux-to-constraint ratio approaches unity ($J/C \rightarrow 1$). In the biological realm, this explains metabolic saturation: as nutrient flow Φ scales up, the mitochondrial and enzymatic constraint C adapts upward, capping the throughput and triggering bottleneck responses (e.g., insulin resistance (chronic C_E upregulation)).

2.2 Notation bridge

Throughout this series, C denotes the operative biological constraint and $\Psi_s = (\Phi/C) \cdot S \cdot M_s$ denotes the extended *adaptive operating ratio*. We use φ for a **transport potential** (chemical, electrical, or mechanical) when gradients appear, to avoid conflating potential with flow intensity Φ .

Status: Derived

2.3 Constitutive relation

For a local transport flux $\mathbf{J}_{\text{flux}}$ driven by potential gradients, a linear resistive closure is

$$\mathbf{J}_{\text{flux}} = -\frac{1}{C} \nabla \varphi. \quad (1)$$

We define scalar **throughput intensity** Φ as an aggregate of transport magnitude appropriate to the subsystem (e.g. pathway flux, perfusion rate, signaling rate). Thus Φ is the biological analogue of “how much is moving,” not the potential φ .

Status: Hypothesis

2.4 Adaptive constraint dynamics (AFL core)

Constraints are not fixed: they stiffen under high throughput and relax when load falls:

$$\frac{dC}{dt} = \alpha \Phi - \beta C, \quad (2)$$

with $\alpha, \beta \geq 0$. This is the AFL **adaptation** clause: C is a state variable coupled to Φ .

Status: Derived (given the postulated ODE)

2.5 Adaptive operating ratio and surface integration

Define the extended operating ratio:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s. \quad (3)$$

We use Ψ_s to represent the fully adaptive state of the biological surface. Interpretive bands (illustrative, not universal thresholds): $\Psi_s < 1$ constrained (tissues under-perfused or restricted); $\Psi_s \approx 1$ balanced (healthy homeostasis ($\Psi_s \approx 1.0$)); $\Psi_s > 1$ overload relative to the chosen scaling. Under this paradigm, disease is interpreted as *surface dysregulation*, where a breakdown in surface responsiveness (S) or rigid accumulation of maladaptive memory (M_s) locks the tissue outside the $\Psi_s \approx 1$ attractor.

Status: Derived

2.6 Assumptions A1–A4

- **A1** A single effective throughput Φ and constraint C capture the phenomenon of interest at the chosen scale.
- **A2** Linear resistance in $\nabla\varphi$ is adequate locally (nonlinear extensions are open).
- **A3** Constraint adaptation follows the two-parameter form (2) with effective (α, β) ; the spatial extension adds diffusion and redistribution as stated in Numerical Verification.
- **A4** Spatial coupling (if used) enters only through conservative redistribution of Φ as specified in the supplement.

Status: Open (domain validity of A1–A4 must be tested per system)

3 Deep Theoretical Foundations: From CDFT to Adaptive Media

3.1 The classical diffusion coefficient and its failure

Standard diffusion is governed by Fick’s first law (Equation 6). The diffusion coefficient D is treated as a material constant — a property of the medium that does not depend on the flux passing through it. This assumption, imported from physics into biology, captures some biological phenomena but fails systematically whenever the medium responds to the flux it carries.

We replace this assumption with a single, far-reaching upgrade.

$$F = \frac{\nabla X}{C(\Omega, x, t)} \quad (4)$$

F = flux; ∇X = thermodynamic driving gradient; $C(\Omega, x, t)$ = adaptive constraint function (system state Ω , position x , time t).

The effective diffusion coefficient becomes:

$$D_{\text{eff}}(\Omega, x, t) = \frac{1}{C(\Omega, x, t)} \quad (5)$$

This is not a cosmetic change. It converts the governing equation from describing a passive, fixed-background medium to describing a **nonlinear adaptive medium** — one in which:

- The medium evolves in response to the flux it carries (C depends on F through Ω)
- Transport is history-dependent (past flux patterns reshape current transport capacity)
- High-flux regions develop different transport properties than low-flux regions
- The system reshapes itself around its own flow patterns

Figure 1 illustrates the contrast between classical ($D = \text{const}$) and adaptive ($D = 1/C(\Omega, x, t)$) diffusion.

Missing figure asset: `figures/fig2_adaptive_medium.png`

Figure 1: **From Passive to Adaptive Medium: The Core CDFT/AFL Upgrade.** (A) Classical diffusion with $D = \text{const}$: transport is symmetric, predictable, and history-independent. The medium is passive. (B) AFL adaptive diffusion with $D = 1/C(\Omega, x, t)$: transport is asymmetric, history-dependent, and nonlinear. The medium reshapes itself. (C) The effective diffusivity $D_{\text{eff}} = 1/C(x, t)$ as a space-time contour: high-flux regions build up constraints (low D , red); relaxed regions remain open (high D , green). The medium has memory. **KEY:** The replacement $D = \text{const} \rightarrow D = 1/C(\Omega, x, t)$ is the single most important upgrade the CDFT/AFL framework makes over classical physics. All other properties of the framework follow from this.

3.2 The constraint function $C(\Omega, x, t)$

The constraint function C is not a fixed material property but a dynamic state variable encoding the system’s current resistance to flux. In biological systems, C depends on:

- **System state Ω :** metabolic configuration, redox balance, energetic capacity, epigenetic state, bioelectric patterning, signalling networks. A mitochondrially depleted cell has high C for energetic processes; a well-oxygenated cell has low C .
- **Flux history F :** constraints respond bidirectionally to flux — both protective upregulation when flux is high and capacity erosion when flux chronically exceeds repair. This is the bidirectional dynamics formalized in Section 4.4.
- **Position x :** constraints vary spatially — a stenosed artery has high C at the stenosis; healthy vascular endothelium has low C ; developing tissues have spatially patterned C governed by bioelectric fields.
- **Time t :** on fast timescales (seconds–hours), C responds acutely to flux changes. On slow timescales (years), C drifts through aging and chronic disease.

3.3 Generality: CDFT subsumes classical transport laws

Classical transport laws are special cases of CDFT under the limiting assumption $C = \text{const}$:

$$\text{Fick's Law: } C = 1/D \Rightarrow F = D \nabla C \quad (6)$$

$$\text{Ohm's Law: } C = 1/\sigma \Rightarrow F = \sigma E \quad (7)$$

$$\text{Darcy's Law: } C = \mu/k \Rightarrow F = (k/\mu) \nabla P \quad (8)$$

$$\text{Newton's viscosity: } C = 1/\nu \Rightarrow F = \nu \nabla v \quad (9)$$

CDFT therefore contains classical transport theory as its passive limit. What it adds is the physics of adaptive media: systems where C itself evolves. Table 1 summarizes the scope across domains.

Table 1: **CDFT Scope: Classical Laws as Passive Special Cases.** All classical transport laws arise from CDFT under the assumption $C = \text{const}$. AFL extends these to the biological adaptive medium where $C = f(\Omega, F, t)$.

Domain	Classical Law	∇X	$C = \text{const}$	AFL	Extension: $C(\Omega, t)$
Biology (general)	AFL/CDFT	Chemical potential / voltage / pressure	Constraint texture	archi-	Adaptive: state, flux, time-dependent $C = f(\text{tissue state})$
Diffusion (physics)	Fick's law	Concentration gradient	$1/D$		
Electrical	Ohm's law	Voltage gradient	$1/\sigma$ (resistance)	C	$= f(\text{channel density})$
Fluid flow	Darcy's law	Pressure gradient	μ/k (viscosity/permeability)	C	$= f(\text{vascular tone})$
Heat	Fourier's law	Temperature gradient	$1/\kappa$ (thermal resistance)	C	$= f(\text{metabolic state})$
Information	Shannon-Hartley	Signal gradient	$1/C$ (channel capacity)	C	$= f(\text{noise, bandwidth})$
Economics	Price dynamics	Price/utility gradient	$1/\text{liquidity}$	C	$= f(\text{market state})$
Ecology	Logistic growth	Resource gradient	$1/K$ (carrying capacity)	C	$= f(\text{ecosystem state})$

4 AFL: The Biological Specification of CDFT

4.1 Biological flux and the system state vector

We define biological flux as $J_i = dS_i/dt$ where S_i is any relevant substrate, signal, or state variable. The system state vector:

$$\Omega = (M, R, E_n, E_p, B_e, S) \quad (10)$$

encodes metabolic configuration M , redox state R , energetic capacity E_n , epigenetic state E_p , bioelectric state B_e , and signalling configuration S . Each component contributes to the adaptive constraint function $C(\Omega, x, t)$ in CDFT. Maximum achievable flux follows the smooth aggregation:

$$J_{\max}(\Omega) = \left[\sum_i J_i(\Omega)^{-\alpha} \right]^{-1/\alpha} \quad (11)$$

In the limit $\alpha \rightarrow \infty$ this recovers $J_{\max} = \min_i \{C_i\}$; for finite α it produces smooth transitions between constraint-dominated regimes.

4.2 From CDFT to the AFL master equation

The constraint function C in CDFT (Equation 4) is specified biologically through constraint stresses:

$$C_i = \frac{f(\Omega)}{C_i(\Omega)} \implies \sigma_i \equiv \frac{J}{C_i(\Omega)} \quad (12)$$

where $C_i(\Omega)$ is the capacity of constraint domain i . Constructing the system instability potential as the weighted sum of squared constraint stresses:

$$V(\Omega) = \sum_i \alpha_i \left[\frac{f(\Omega)}{C_i(\Omega)} \right]^2 \quad (13)$$

and applying the CDFT gradient-descent principle $d\Omega/dt = -\nabla V(\Omega)$, we obtain the AFL master equation:

$$\frac{d\Omega}{dt} = -\nabla \left[\sum_i \alpha_i \left(\frac{f(\Omega)}{C_i(\Omega)} \right)^2 \right] \quad (14)$$

Ω = system state (Eq. 10); $f(\Omega)$ = biological flux; $C_i(\Omega)$ = constraint capacities; α_i = biological importance weights. This is CDFT (Eq. 4) with $C = \sum_i \alpha_i [f(\Omega)/C_i(\Omega)]^2$.

4.3 Lyapunov stability argument

Theorem 1 (AFL Stability). *The equilibria Ω^* of the AFL dynamical system are Lyapunov stable, and trajectories converge to $\{\Omega : \nabla V(\Omega) = 0\}$ as $t \rightarrow \infty$.*

Argument. $V(\Omega) \geq 0$ for all Ω (sum of squared terms), with $V(\Omega^*) = 0$. Computing the time derivative:

$$\frac{dV}{dt} = \nabla V \cdot \frac{d\Omega}{dt} = \nabla V \cdot (-\nabla V) = -\|\nabla V\|^2 \leq 0 \quad (15)$$

By Lyapunov's theorem [Lyapunov, 1992], equilibria Ω^* satisfying $\nabla V(\Omega^*) = 0$ are stable. If V is radially unbounded, LaSalle's invariance principle ensures convergence. $\square$

4.4 The recursive Ω - J loop and timescale separation

Flux modifies the constraint function C through the recursive coupling:

$$\Omega_{t+1} = F(\Omega_t, f(\Omega_t)) \quad (16)$$

This self-referential loop — state generates flux, flux modifies C , modified C reshapes future flux — is the mechanism by which biological media adapt. In CDFT language: the flux carried by the medium gradually changes the medium's own transport coefficients.

Timescale separation governs how quickly this occurs:

$$\frac{dJ}{dt} = f(\Omega, J_{\text{req}}) \quad [\text{fast: seconds--minutes; flux adaptation}] \quad (17)$$

$$\frac{d\Omega}{dt} = \varepsilon \cdot F(\Omega, J), \quad 0 < \varepsilon \ll 1 \quad [\text{slow: hours--generations; structural drift}] \quad (18)$$

The coupling loop $B_e \rightarrow S \rightarrow E_p \rightarrow \text{Enz} \rightarrow J \rightarrow \Omega \rightarrow B_e$ governing adaptation, homeostasis ($\Psi_s \approx 1.0$), disease, and developmental coordination is illustrated in Figure 2.

Missing figure asset: `figures/fig3_omega_loop.png`

Figure 2: **The AFL Ω - J Recursive Loop and Timescale Hierarchy.** (A) The closed regulatory cycle $B_e \rightarrow S \rightarrow E_p \rightarrow \text{Enz} \rightarrow J \rightarrow \Omega \rightarrow B_e$. In CDFT terms: flux modifies C , which modifies future flux capacity. (B) Timescale separation: flux adapts fast ($\varepsilon \ll 1$); structural state Ω evolves slowly. Chronic constraint exposure produces structural shifts that persist after the acute constraint resolves. **KEY:** The adaptive medium *remembers* its flux history through Ω . This history-dependence is what makes biological transport fundamentally different from classical diffusion.

5 Thermodynamic Foundation

5.1 Biological systems as nonequilibrium adaptive media

Living systems operate far from thermodynamic equilibrium [Schrodinger, 1944, Prigogine and Stengers, 1984]. In linear nonequilibrium thermodynamics [Onsager, 1931, Prigogine, 1967], flux arises from thermodynamic forces:

$$J_i = L_i X_i \quad (19)$$

where L_i is a transport coefficient and X_i is the conjugate thermodynamic driving force. In CDFT language: $L_i = 1/C_i$. The key insight is that in biological systems, L_i is not a material constant — it is a dynamical variable that responds to flux, state, and history.

Entropy production:

$$\dot{S} = \sum_i J_i X_i \geq 0 \quad (20)$$

defines the thermodynamic cost of maintaining biological organization. The structural limits on this entropy export translate into AFL constraint capacities. Living systems maintain internal order by exporting entropy — and it is the architecture of this export machinery that defines C .

5.2 The Nernst grounding and bioelectric constraints

The bioelectric component of Ω provides the clearest example of a thermodynamically grounded constraint:

$$E = \frac{RT}{zF} \ln \left(\frac{[\text{ion}]_{\text{out}}}{[\text{ion}]_{\text{in}}} \right) \quad (21)$$

Ion gradients represent stored free energy driving electrical signalling, membrane transport, and tissue-level coordination [Levin, 2021, 2012]. In CDFT terms, the membrane potential sets C_{electric} : a depolarized membrane has high C for ion-dependent signalling; a hyperpolarized membrane has low C and high transport efficiency.

5.3 Maximum entropy and the AFL potential

The AFL potential $V(\Omega)$ admits interpretation as constraint stress — the degree to which the system’s free energy processing capacity is being utilized. Connection to maximum entropy production [Martyushev and Seleznev, 2006]: while thermodynamics drives systems toward higher flux along free energy gradients, the AFL constraint architecture defines the ceiling J_{max} that limits this

drive. The master equation describes the biological system's trajectory as it maximizes entropy production subject to its constraint architecture.

6 The Dimensionless Constraint Utilization Index Φ

6.1 Φ as the AFL Reynolds number

For each constraint domain:

$$\Phi_i = \frac{J}{C_i(\Omega)} \equiv \frac{F}{C_i^{-1}} = F \cdot C_i \quad (22)$$

Φ is the AFL analogue of the Reynolds number in fluid dynamics: a dimensionless predictor of regime transitions. The system-level index $\Phi_{\text{sys}} = \max_i \Phi_i$. At $\Phi = 1$ (the constraint boundary), the system undergoes a pitchfork bifurcation:

$$\frac{dx}{dt} = a(1 - \Phi)x - bx^3 \quad (23)$$

For $\Phi < 1$: stable functional states. For $\Phi > 1$: only collapsed state remains. This is the mathematical signature of sudden clinical decompensation after prolonged apparent stability.

Missing figure asset: `figures/fig4_phi_bifurcation.png`

Figure 3: **The AFL Dimensionless Index Φ and Phase Transition at $\Phi = 1$.** **(A)** The Φ gauge. In CDFT language: Φ measures how close the flux is to the constraint ceiling $1/C$. $\Phi = 0$ means $C \rightarrow 0$ (open medium); $\Phi = 1$ means C has risen to match the driving gradient; $\Phi > 1$ means the constraint ceiling has been exceeded (system in violation). **(B)** Pitchfork bifurcation at $\Phi = 1$: stable functional states exist for $\Phi < 1$; at $\Phi = 1$ a phase transition occurs; only collapsed state $x^* = 0$ remains for $\Phi > 1$. **KEY:** Φ is the AFL analogue of Reynolds number (Re) in fluid dynamics and R_0 in epidemiology — a single dimensionless number predicting regime transitions.

7 Main Results

7.1 AFL definition

Adaptive Flux Limitation is the joint dynamics of throughput and constraint such that C co-evolves with Φ according to (2), producing regulable Ψ_s . Equation (2) is the **0D AFL closure**; spatial extensions add redistribution terms as in the supplement.

Status: Derived

7.2 Bottleneck (informal)

When multiple serial constraints C_i limit throughput, the effective ceiling is often set by the tightest link; a minimal representation is $\Phi_{\text{sys}} \lesssim \min_i C_i$ in appropriate units.

Status: Inferred

7.3 Saturation

As external drive increases, measured throughput may plateau because C rises with load or because Φ approaches a hard capacity; both are consistent with AFL-style regulation.

Status: Inferred

8 Numerical Verification

The deterministic numerical check is a 2D lattice toy model with fixed seed 42. A public supplementary script should be released under active paper numbering; the model itself is specified by conservative redistribution of Φ/C and constraint update

$$\partial_t C = \alpha |\text{redistribution}| - \beta C + \gamma \nabla^2 C, \quad (24)$$

discretized explicitly (details in-file). Tables T1–T3 report: final mean Ψ_s with AFL on vs. $\alpha = 0$ control; regime labels; spatial standard deviation of C as a proxy for channeling.

Status: Derived

9 Interpretation

Metabolic, vascular, and signaling examples fit the pattern “drive $\rightarrow$ throughput $\rightarrow$ adaptive surface integration,” but each domain requires its own measurement map from observables to the 4-state variables (Φ, C, S, M_s) . Diseases may present as persistent mismatch driven by surface dysregulation: elevated C (fibrosis (maladaptive C elevation), inflammation (system-wide constraint C amplification)), insufficient Φ (ischemia (severe Φ collapse)), or runaway Ψ_s in cytokine storm-like caricatures driven by unconstrained memory feedback.

Status: Inferred

10 Limitations and Open Problems

- Single-field reductions omit compartmental structure and control loops.
- Parameters (α, β, γ) are not identified from data here.
- Bands for Ψ_s are interpretive, not universal diagnostic cutoffs.

Status: Open

11 Clinical Mapping of Symbols

Symbol	Clinical meaning	Measurement proxy	Invalidation condition
Φ	Physiological flux intensity	Substrate flux, perfusion rate, signaling rate	If Φ cannot be distinguished from driving gradient $\nabla\varphi$, mapping is not useful
C	Constraint / pathway resistance	Dynamic: lab markers of resistance; static: enzyme capacity	If C does not respond to Φ changes (i.e., $\alpha \approx 0$), AFL adds nothing over static models
S	Surface responsiveness	Tissue adaptation score; receptor density index	If S is constant across all patients, it can be absorbed into scaling
M_s	Surface memory / structural hysteresis	Biomarker of locked state: fibrosis score, epigenetic age	If M_s relaxes to baseline instantly, memory framing is unnecessary
Ψ_s	Operating state ratio	Computed from pairs of flux/constraint markers; not a direct assay	If Ψ_s is indistinguishable from a simple Φ/C ratio without $S \cdot M_s$, the extended model adds no value

Status: Hypothesis

12 Known Medicine Baseline

AFL is layered on top of established physiology, not a replacement for it:

- Adaptive regulation is well established: insulin secretion adapts to chronic glycaemic load; the kidney adjusts filtration and tubular function with sodium load; the heart remodels under chronic pressure overload.
- AFL proposes a common mathematical language for these adaptations, not a new mechanism.
- The $\Psi_s \approx 1$ homeostatic attractor is an interpretive framework, not a clinically validated threshold.
- When AFL framing differs from standard clinical interpretation, standard clinical interpretation takes precedence.

13 Experiments and Future Validation

1. **Mathematical consistency:** finite Ψ_s at all parameters; stable equilibrium check; sensitivity to α, β, γ .
2. **Synthetic perturbation:** release-local script (`supplementary/supplementary_afl_01.py`) — adaptive vs control 2-D lattice; metabolic channeling signature.

3. **In vitro:** perfused tissue challenge — does adaptive constraint formation produce channel-like heterogeneity in a controlled perfusion assay?
4. **Ex vivo:** organ-on-chip flow challenge — repeated load/recovery cycle; measurement of spatial constraint heterogeneity.
5. **Retrospective:** longitudinal biomarker datasets — does constraint marker elevation precede flux marker decline?
6. **Prospective:** cell atlas mapping of candidate $C(x, t)$ regions using HuBMAP spatial data.

Falsification: if challenge response is fully explained by a single static concentration without adaptive constraint dynamics, AFL adds no mechanistic value.

14 Clinical Safety Boundary

This paper establishes a notation framework. It does not provide diagnostic thresholds, treatment guidance, or clinical decision tools. Ψ_s values computed from the toy model are not patient measurements.

15 Runtime Diagnostic

Release-local script: `supplementary/supplementary_afl_01.py`. Outputs in `outputs/paper_01/afl_scenarios.csv`, `surface_response.csv`, `adaptive-vs-control.png`, `summary.json`. All outputs are toy diagnostics. Claim status: mathematical demonstration only.

16 Conclusion

AFL-1 reframes a slice of physiology as coupled throughput–constraint dynamics governed by the adaptive operating ratio $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The toy model supports a minimal spatial demonstration of adaptive channeling vs. non-adaptive control, illustrating how biological surfaces dynamically integrate load. Clinical interpretation requires domain-specific measurement maps and validation.

Status: Inferred

17 Relation to the CFD physics manuscripts

The CFD physics manuscripts and the OOL rewrite track share the same **claim discipline**: explicit assumptions, separation of derivation vs. interpretation, and script-backed numerics. AFL applies the notation layer $(\Phi, C, S, M_s, \Psi_s)$ to biology and clinic; it does not assert new vacuum or particle predictions.

Status: Inferred

18 Reproducibility Note

- **Script:** `supplementary/supplementary_afl_01.py` (NumPy + matplotlib; seed 42; runtime < 2s).
- **Outputs:** `outputs/paper_01/` — CSV tables, PNG figure, summary JSON.

Status: Derived

Conflict of Interest

The author declares no conflict of interest.

Funding

This research received no external funding.

Author Contributions

Conceptualization, formal analysis, runtime engineering, and manuscript preparation: S.B.M.

Data Availability

All computational models, Python scripts, and numerical verifications are available in the **supplementary/** directory.

19 Reference Layer

References

- Michael Levin. Molecular bioelectricity in developmental biology: New tools and recent discoveries. *BioEssays*, 34(3):205–217, 2012. doi: 10.1002/bies.201100136.
- Michael Levin. Bioelectric signaling: Reprogrammable circuits underlying embryogenesis, regeneration, and cancer. *Cell*, 184:1971–1989, 2021.
- A. M. Lyapunov. *The General Problem of the Stability of Motion*. Taylor and Francis, 1992.
- Leonid M. Martyushev and Vladimir D. Seleznev. Maximum entropy production principle in physics, chemistry and biology. *Physics Reports*, 426(1):1–45, 2006. doi: 10.1016/j.physrep.2005.12.001.
- Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven flux dynamics. 2026. doi: 10.5281/zenodo.20250821. URL <https://doi.org/10.5281/zenodo.20250821>.
- Lars Onsager. Reciprocal relations in irreversible processes. i. *Physical Review*, 37(4):405–426, 1931. doi: 10.1103/PhysRev.37.405.
- Ilya Prigogine. *Introduction to Thermodynamics of Irreversible Processes*. Interscience, 3 edition, 1967.
- Ilya Prigogine and Isabelle Stengers. *Order Out of Chaos: Man’s New Dialogue with Nature*. Bantam Books, 1984.
- Erwin Schrodinger. *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, 1944.